

Introduction to Number Theory

Number theory is the branch of mathematics concerned with integers. Traditionally, number theory was a pure branch of mathematics—known for its abstract nature rather than its applications.

5.1 ♦ Divisors

Let n and d be integers, $d \neq 0$. We say that d divides n if there exists an integer q satisfying $n = dq$. We call q the quotient and d a divisor or factor of n . If d divides n , we write $d \mid n$. If d does not divide n , we write $d \nmid n$.

Example 5.1.2 Since $21 = 3 \cdot 7$, 3 divides 21 and we write $3 \mid 21$. The quotient is 7. We call 3 a divisor or factor of 21.

Theorem 5.1.3 Let m , n , and d be integers.

(a) If $d \mid m$ and $d \mid n$, then $d \mid (m + n)$.

(b) If $d \mid m$ and $d \mid n$, then $d \mid (m - n)$.

(c) If $d \mid m$, then $d \mid mn$.

Definition 5.1.4 An integer greater than 1 whose only positive divisors are itself and 1 is called **prime**. An integer greater than 1 that is not prime is called **composite**.

Example 5.1.6 By inspection, we find that none of 2, 3, 4, 5, ..., 41, 42 divides 43; thus, 43 is **prime**. Checking the list 2, 3, 4, 5, ..., 449, 450 for potential divisors of 451, we find that 11 divides 451 ($451 = 11 \cdot 41$); thus, 451 is **composite**.

Theorem 5.1.7 A positive integer n greater than 1 is composite if and only if n has a divisor d satisfying $2 \leq d \leq \sqrt{n}$.

Algorithm 5.1.8

Testing Whether an Integer is Prime

This algorithm determines whether the integer $n > 1$ is prime. If n is prime, the algorithm returns 0. If n is composite, the algorithm returns a divisor d satisfying $2 \leq d \leq \sqrt{n}$. To test whether d divides n , the algorithm checks whether the remainder when n is divided by d , $n \bmod d$, is zero.

Input: n

Output: d

is prime(n)

{for $d = 2$ to $[\sqrt{n}]$ if ($n \bmod d == 0$)

return d

return 0

}

Example 5.1.9 To determine whether 43 is prime, Algorithm 5.1.8 checks whether any of 2, 3, 4, 5, $6 = \lceil \sqrt{43} \rceil$ divides 43. Since none of these numbers divides 43, the condition

$$n \bmod d == 0 \quad (5.1.5)$$

is always false.

Therefore, the algorithm returns 0 to indicate that 43 is prime.

To determine whether 451 is prime, Algorithm 5.1.8 checks whether any of 2, 3, ..., 21 = $\lceil \sqrt{451} \rceil$ divides 451. For $d = 2, 3, \dots, 10$, d does not divide 451 and the condition (5.1.5) is false. However, when $d = 11$, d does divide 451 and the condition 5.1.5 is true. Therefore, the algorithm returns 11 to indicate that 451 is composite and 11 divides 451.

Theorem 5.1.11

Fundamental Theorem of Arithmetic

Any integer greater than 1 can be written as a product of primes. Moreover, if the primes are written in nondecreasing order, the factorization is unique. In symbols, if

$$n = p_1 p_2 \cdots p_i,$$

where the p_k are primes and

$$p_1 \leq p_2 \leq \cdots \leq p_i, \text{ and}$$

$$n = p'_1 p'_2 \cdots p'_j,$$

where the p'_k are primes and

$$p'_1 \leq p'_2 \leq \cdots \leq p'_j,$$

then $i = j$

and $p_k = p'_k$ for all $k = 1, \dots, i$.

Theorem 5.1.12 The number of primes is infinite.

Example 5.1.13 Show how the proof of Theorem 5.1.12 produces a prime larger than 11.

SOLUTION We list the primes less than or equal to 11: 2, 3, 5, 7, 11. We let $m = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 + 1 = 2311$. Using Algorithm 5.1.8, we find that 2311 is prime. We have found a prime, namely 2311, larger than each of 2, 3, 5, 7, 11. (If 2311 had turned out not to be prime, Algorithm 5.1.8 would have found a factor of 2311, which would necessarily be larger than each of 2, 3, 5, 7, 11.)

The **greatest common divisor** of two integers m and n (not both zero) is the largest positive integer that divides both m and n . For example, the greatest common divisor of 4 and 6 is 2, and the greatest common divisor of 3 and 8 is 1.

Definition 5.1.14 Let m and n be integers with not both m and n zero. A common divisor of m and n is an integer that divides both m and n . The greatest common divisor, written $\gcd(m, n)$, is the largest common divisor of m and n .

Example 5.1.15 The positive divisors of 30 are

1, 2, 3, 5, 6, 10, 15, 30,

and the positive divisors of 105 are

1, 3, 5, 7, 15, 21, 35, 105;

thus the positive common divisors of 30 and 105 are

1, 3, 5, 15.

It follows that the greatest common divisor of 30 and 105,

$\gcd(30, 105)$, is 15.

Example 5.1.16 We find the greatest common divisor of 30 and 105 by looking at their prime factorizations

$$30 = 2 \cdot 3 \cdot 5$$

$$105 = 3 \cdot 5 \cdot 7.$$

Notice that 3 is a common divisor of 30 and 105 since it occurs in the prime factorization of both numbers. For the same reason, 5 is also a common divisor of 30 and 105. Also, $3 \cdot 5 = 15$ is also a common divisor of 30 and 105. Since no larger product of primes is common to both 30 and 105, we conclude that 15 is the greatest common divisor of 30 and 105.

Theorem 5.1.17

Let m and n be integers, $m > 1$, $n > 1$, with prime factorizations

$$m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

and

$$n = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}.$$

(If the prime p_i is not a factor of m , we let $a_i = 0$. Similarly, if the prime p_i is not a factor of n , we let $b_i = 0$.)

Then

$$\gcd(m, n) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_k^{\min(a_k, b_k)}.$$

Example 5.1.18 Using the notation of Theorem 5.1.17, we have

$$82320 = 2^4 \cdot 3^1 \cdot 5^1 \cdot 7^3 \cdot 11^0$$

and

$$950796 = 2^2 \cdot 3^2 \cdot 5^0 \cdot 7^4 \cdot 11^1.$$

By Theorem 5.1.17,

$$\gcd(82320, 950796) =$$

$$2^{\min(4,2)} \cdot 3^{\min(1,2)} \cdot 5^{\min(1,0)} \cdot 7^{\min(3,4)} \cdot 11^{\min(0,1)} = 2^2 \cdot 3^1 \cdot 5^0 \cdot 7^3 \cdot 11^0 = 4116.$$

Neither the “list all divisors” method of Example 5.1.15 nor use of prime factorization as in Example 5.1.18 is an efficient method of finding the greatest common divisor.

The problem is that both methods require finding the prime factors of the numbers involved and no efficient algorithm is known to compute these prime factors.

A companion to the greatest common divisor is the least common multiple.

Definition 5.1.19 Let m and n be positive integers. A common multiple of m and n is

an integer that is divisible by both m and n . The least common multiple, written $\text{lcm}(m,n)$, is the smallest positive common multiple of m and n .

Example 5.1.20

The least common multiple of 30 and 105, $\text{lcm}(30,105)$, is 210 because 210 is divisible by both 30 and 105 and, by inspection, no positive integer smaller than 210 is divisible by both 30 and 105.

Example 5.1.21

We can find the least common multiple of 30 and 105 by looking at their prime factorizations

$$30 = 2 \cdot 3 \cdot 5 \qquad 105 = 3 \cdot 5 \cdot 7.$$

The prime factorization of $\text{lcm}(30,105)$ must contain 2, 3, and 5 as factors [so that 30 divides $\text{lcm}(30,105)$]. It must also contain 3, 5, and 7 [so that 105 divides

$\text{lcm}(30,105)]$. The smallest number with this property is

$$2 \cdot 3 \cdot 5 \cdot 7 = 210.$$

Therefore, $\text{lcm}(30,105) = 210$.

Theorem 5.1.22

Let m and n be integers, $m > 1$, $n > 1$, with prime factorizations

$$m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

and

$$n = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}.$$

(If the prime p_i is not a factor of m , we let $a_i = 0$. Similarly, if the prime p_i is not a factor of n , we let $b_i = 0$.) Then

$\text{lcm}(m,n) =$

$$p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_k^{\max(a_k, b_k)}.$$

Example 5.1.23

Using the notation of Theorem 5.1.22, we have

$$82320 = 2^4 \cdot 3^1 \cdot 5^1 \cdot 7^3 \cdot 11^0$$

and

$$950796 = 2^2 \cdot 3^2 \cdot 5^0 \cdot 7^4 \cdot 11^1.$$

By Theorem 5.1.22,

$$\begin{aligned} \text{lcm}(82320, 950796) &= \\ 2^{\max(4,2)} \cdot 3^{\max(1,2)} \cdot 5^{\max(1,0)} \cdot 7^{\max(3,4)} \cdot 11^{\max(0,1)} &= \\ 2^4 \cdot 3^2 \cdot 5^1 \cdot 7^4 \cdot 11^1 &= 19015920. \end{aligned}$$

Example 5.1.24

In Example 5.1.15, we found that $\gcd(30, 105) = 15$, and in Example 5.1.21, we found that $\text{lcm}(30, 105) = 210$. Notice that the product of the gcd and lcm is equal to

the product of the pair of numbers; that is,

$$\gcd(30,105) \cdot \text{lcm}(30,105)$$

$$= 15 \cdot 210 = 3150 = 30 \cdot 105.$$

Theorem 5.1.25

For any positive integers m and n ,

$$\gcd(m,n) \cdot \text{lcm}(m,n) = mn.$$